

Yang Liu

✉ liuyang3489xyz@gmail.com ☎ 3463390270 📍 1885 EL Paseo St, APT 346, Houston-TX 77054

🌐 [linkedin.com/in/yang-liu-278658195](https://www.linkedin.com/in/yang-liu-278658195) 🔄 <https://yliu38.github.io/> (click here for personal page)

Education

- PhD in Biostatistics and Data Science**, *University of Texas Health Science Center at Houston* 08/2021 – 08/2025
Houston, United States
- Relevant Courses: Statistical Inference, Machine Learning, Survival Analysis, Clinical Trials
 - Honors: Caroline Ross Fellowship, Julius and Suzan Glickman Endowed Scholarship.
- Master in Biomedical Informatics**, *University of Texas Health Science Center at Houston* 01/2019 – 08/2021
Houston, United States
- Relevant Courses: Big Data in Biomedical Informatics, Stat Analysis of Genomic Data
 - Honors: Paul C. Boyle Research Award (×2), Dean's Excellence Scholarship, Schull Institute Summer Program Scholarship, James Turley Endowed Scholarship.
- Master in Cancer Biology**, *Seoul National University* 08/2015 – 08/2017
Seoul, South Korea
- Relevant Courses: Medical Statistics, Molecular Cell Biology
 - Honors: SNU Global Scholarship (×2)

Professional Experience

- Graduate Research Assistant**, *University of Texas MD Anderson Cancer Center* 01/2020 – 08/2025
Houston, United States
- **Engineered a novel deep learning framework** combining network-based feature engineering with **graph neural networks (GNNs)** to classify cancer phenotypes and identify biomarkers from gene expression data; secured \$519K Moon Shot funding (bioRxiv, 2025 📄)
 - **Developed a network-penalized statistical method** for biomarker identification in gene expression datasets with survival endpoints (bioRxiv, 2025 📄)
 - **Integrated scRNA-seq and scATAC-seq datasets** to elucidate functional consequences of germline risk variants in glioma progression; awarded \$461K R01 funding 📄
 - **Enhanced ICD-10 code prediction** from clinical notes by leveraging data-derived knowledge graphs and fine-tuned lightweight large language models (**LLMs**)
 - **Developed a RAG tool** to answer graduation-related queries at UTHealth for personalized student support (Huggingface spaces 📄)
 - **Built multimodal transformer models** for segmentation of Organ-at-Risk (OAR) regions using paired MRI and CT images (Github 📄)
- R&D Engineer**, *Berry Genomics* 11/2017 – 07/2018
Beijing, China
- Developed a BRCA diagnostic kit on NGS platform; optimized primer design for coverage
 - Coordinated cross-departmental collaboration for product development

Academic Projects

Microsatellite Status Prediction
Achieved comparable **CNN** performance for classification of histological images with limited GPU resources

Meta-analysis of mortality rate of COVID-19
Implemented **Bayesian** hierarchical models for mortality estimation

Batch-effect Correction for Single-cell RNA Sequencing Data
Compared 7 methods including Seurat and Harmony for single-cell RNA sequencing data integration

- Selected course projects**
- Built a **Yelp scraper** to rank nearby Asian restaurants using a custom preference score
 - **Imputed missing data** using a custom **EM** algorithm with accuracy matching standard tools.
 - Designed a **Markov chain model** to predict outcomes in bone marrow transplant patients
 - Developed **optimization algorithms** using coordinate descent and ADMM techniques
 - Applied **NLP** methods (word embeddings, random forest) to classify spam emails

Publications & Conferences

- **Authored 6 abstracts** published in *Neuro-Oncology* and *Molecular Cancer Therapeutics*, **2 first-authored peer-reviewed publications**, and **3 first-authored preprints (2025) currently under review**
- **Presented posters** at SNO Annual Meetings (2021, 2024) and Brain Tumor Center Retreat (2023)
- **Presented at** FY22 Brain Tumor Center Retreat, GraphConnect 2022, and Neo4j AWS GenAI Lab 2024

Skills & Interests

- **Tools:** R, Python, SQL, Neo4j (Cypher), SAS, Tableau, Linux, HPC, CUDA, Parallel Computing
- **Specializations:** Statistics, AI/ML, Bioinformatics, Bayesian, Simulation, Graph and Image Analysis, LLMs, Wet Lab
- **Languages:** Chinese (Native), English (Proficient), Korean (Proficient)
- **Interests:** Swimming, Hiking, Piano